

MARRI LAXMAN REDDY INSTITUTE OF TECHNOLOGY

Department of Aeronautical Engineering

AI Knowledge Base Document

1. Department Overview

The Department of Aeronautical Engineering at MLRIT provides excellent infrastructure and state-of-the-art laboratories, creating an ideal environment for research and innovation in one of the most challenging branches of engineering.

The dedicated faculty and passionate students share a common vision of connecting the fastest-growing aerospace sector to the economic growth of the nation.

Key Highlights

- Accredited by the National Board of Accreditation (NBA) under Tier-I in 2019.
- Student-centric learning approach with unique teaching methodologies: Micro-Projects, COTS, EPICS, SPEED.
- Students have received Gold Medals twice at the university level.
- Four Centres of Excellence established in Product Lifecycle Management, Flight Simulation, Digital Manufacturing, and Computational Engineering.
- Faculty and students have published approximately 160 papers in Scopus-indexed journals, plus two books in core subjects.
- Students have won cash prizes totalling Rs. 5 Lakhs in the last 3 years.
- 10 patents published in 2019-2020.
- Handling Rs. 50 Lakh worth DST-funded projects under FIST and Rs. 17 Lakh AICTE project.
- Received funding from ISRO and SERB to organise international conferences ICAAMM-2016 and ICAAMM-2018.
- Well-established laboratories with state-of-the-art equipment and licensed software.
- One of only seven institutions in India selected for the sponsored Boeing Externship for III year students.
- Offers a special course to employees of Tata Advanced Systems Limited (TASL).
- Tata Technologies provides in-house training in PLM and recruits trained students.
- CATIA Certification Centre by Dassault Systemes.

- In-house Campus Recruitment Training (CRT) programme throughout the year.
- Professional Associations: AeSI, SAE, ISTE, and IAAA.
- Students have participated in SAE International competition at Florida, USA.
- Regularly organises conferences, FDPs, workshops, guest lectures, and industrial visits.

2. Vision and Mission

Vision

To be a centre of excellence in Aeronautical Engineering with emphasis on Research and Innovation to serve the needs of industry with human values to build a strong nation.

Mission

Consistently produce top-quality Aeronautical engineers with core and multidisciplinary skills, who can become ace leaders and successful entrepreneurs with human values.

3. Key Differentiators

- First ever in South India: Centre of Excellence in Product Life Cycle Management, established in collaboration with Tata Technologies.
- One of only seven institutions in India sponsored by Boeing for the Externship Programme for III year students.
- The only CATIA Certification Centre by Dassault Systems in Andhra Pradesh and Telangana.

4. MoUs (Memorandums of Understanding)

Organisation	Objective of MOU
Boeing	Two months paid externship (Rs. 10,000 per month) to two students of III B.Tech II Semester. Boeing team visits the campus once every three years to interact with students.
AerX Labs India Pvt. Limited	Provides assistance to establish Centre of Excellence for Flight Simulation Lab. Provides assistance to carry out student projects in the field of aviation.
Dassault Systemes	CATIA Certification centre. Provides paid internships to students.

CSIR-NAL	Helps to establish Composite and Flight Simulation lab. Provides internships to M.Tech students.
Tata Advanced Systems Limited (TASL)	MLRIT provides special diploma course for TASL employees. TASL provides internships to MLRIT students. TASL visits MLRIT to recruit Aero and Mechanical students.
Tata Technologies	Establishes the PLM Lab and provides training to students. Total expenditure shared equally between Tata Technologies and MLRIT (50% + 50%).

5. Career Prospects

Public Sector

- HAL
- NAL
- DRDO
- ISRO
- ADA

Aerospace Industries

- Boeing
- Airbus
- BE Aerospace
- GE
- Rolls Royce
- Honeywell

Service Sectors

- Cyient
- TASL
- VEM Technologies
- NUCON Aerospace

Aviation and Maintenance

- Mphasis
- Capgemini
- Camp Systems
- GMR Aerospace

Higher Education

- M.Tech
- MS in India and abroad

6. Program Outcomes (POs)

Engineering graduates of the Aeronautical Engineering programme will be able to:

Code	Description
PO1	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
PO4	Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO5	Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities, with an understanding of their limitations.
PO6	The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	Environment and Sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics, responsibilities, and norms of engineering practice.
PO9	Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, including writing effective reports and design documentation, making presentations, and giving and receiving clear instructions.

PO11	Project Management and Finance: Demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects in multidisciplinary environments.
PO12	Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

7. Program Specific Outcomes (PSOs)

Code	Description
PSO1	Apply engineering and management knowledge and techniques to estimate time and resources needed to complete Aerospace/Mechanical engineering projects.
PSO2	Recognize the challenging and rewarding careers in the field of Aerospace Engineering.

8. Laboratories

Engineering Workshop Lab	
Purpose	Provides introduction to basic manufacturing techniques used in industry. Also used to manufacture components such as Airfoil, cylinder, and scaled wing models for wind tunnel testing.
Facilities	Bench Vices, Arc Welding equipment, Gas Welding equipment, Lathes, and other machining processes. Display boards provided for equipment list, experiments, timetables, and syllabus.
Used By	I Year, I Semester

Mechanics of Solids Lab	
Purpose	Introduces students to basic properties of structural materials and behavior of structural elements under loading conditions. Experiments measure material properties such as impact strength, tensile strength, compressive strength, hardness, and ductility.
Facilities	Universal Testing Machine, Torsion Testing Machine, Impact Testing Machine, Brinell Hardness Testing Machine, Rockwell Hardness Testing Machine. White marker board and display boards provided.
Used By	II Year, I Semester

Mechanics of Fluids Lab

Purpose	Introduces mechanics of incompressible and compressible fluids. Covers measurement of flow rates, capacity, and velocity of water in tanks, closed pipes, and open channels. Covers principles and important theories of modern fluid mechanics.
Facilities	Venturi meter, Orifice meter, Reciprocating pump, Multi-stage reciprocating pump. White marker board and display boards provided.
Used By	II Year, I Semester

Aircraft Production Technology-I Lab

Purpose	Focuses on basic manufacturing processes. Students perform experiments converting raw materials into useful jobs with given dimensions and shapes using various manufacturing techniques.
Facilities	Pattern Making machines, Sand Moulding, Injection Moulding, Arc Welding, Gas Welding equipment. White marker board and display boards provided.
Used By	II Year, I Semester

Computer Aided Drafting Lab

Purpose	Introduces the drawing and design process. Covers basic principles of aircraft design and all design techniques. Students gain knowledge to design different parts of an aircraft including machine components, sub-assemblies, and main assemblies.
Facilities	AutoCAD software installed on all systems. White marker board and display boards provided.
Used By	II Year, II Semester

Aircraft Production Technology-II Lab

Purpose	Focuses on various manufacturing processes. Students perform experiments converting raw materials into useful jobs with given dimensions and shapes using various manufacturing techniques.
Facilities	Lathe machine, Milling machine, Shaping machine, and other conventional machines. White marker board and display boards provided.
Used By	II Year, II Semester

Aerodynamics Lab

Purpose	Conducts studies in fundamental aerodynamics and propulsion. Covers passive and active flow control techniques, prediction and measurement of forces and moments on bodies moving through a fluid. Research includes
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	viscous drag reduction, energy efficiency, high-speed jet flows, aero acoustics, turbo machinery, and combustion aero acoustics.
Facilities	Display boards provided for equipment list, experiments, timetables, and syllabus.
Used By	II Year, II Semester

Aerospace Structure Lab

Purpose	Enhances undergraduate learning by encouraging project work. Provides facilities for specimen preparation and testing of structural components.
Facilities	Maxwell's Reciprocal Theorem apparatus, Universal Testing Machine, Shear Center apparatus, test frames with actuators for testing various structural components. White marker board and display boards provided.
Used By	III Year, I Semester

Propulsion Lab

Purpose	Provides practical understanding of Fluid Mechanics, Thermodynamics, and Propulsion. Students explore sources of propulsive forces, performance parameters for air-breathing propulsion systems and rockets, and principles of 2-stroke and 4-stroke engines.
Facilities	2-stroke and 4-stroke engine setups, petrol and diesel engine test setups. Display boards provided for equipment list, experiments, timetables, and syllabus.
Used By	III Year, I Semester

Flight Simulation Lab

Purpose	Deals with real-time problems of Flight Data Analytics. Students acquire basic knowledge of pilot training through virtual environment simulation. The lab has a flight simulator setup with live shares.
Facilities	Flight Simulator with live shares. Display boards provided for experiments, timetables, and syllabus.
Used By	III Year, I Semester

Flight Vehicle Design Lab

Purpose	Deals with design and weight estimation of aircraft based on the Request for Proposal (RFP). Covers different modes of design and evaluation of empty weight fraction and payload.
Facilities	MS-Excel for iterative payload calculations, White marker board, LCD projector. Display boards provided for experiments, timetables, and syllabus.

Used By	III Year, II Semester
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Aircraft Interior Design Lab

Purpose	Introduces overview of drawing and design process for aircraft. Students gain knowledge to design individual components, sub-assemblies, and main assemblies, and study the different phases of the design process.
Facilities	CATIA V5 software and AutoCAD installed on all systems. White marker board and display boards provided.
Used By	III Year, II Semester

Computational Fluid Dynamics Lab

Purpose	Requires understanding and practice of ANSYS Fluent and MATLAB commercial software packages. Fluid dynamics analysis is performed using ANSYS Fluent; code development experiments use MATLAB. Students carry out all project work related to these software packages.
Facilities	ANSYS Fluent software, MATLAB software, all required software installed on systems. White marker board and display boards provided.
Used By	IV Year, I Semester

Heat Transfer Lab

Purpose	Experiments help students understand all three modes of heat transfer: conduction, convection, and radiation. Experiments also develop innovative skills and serve as a starting point for higher-level experimental studies in heat transfer.
Facilities	Experimental setups for conduction, convection, and radiation. White marker board and display boards provided.
Used By	IV Year, I Semester

Aero Materials Lab

Purpose	Used to analyse surface conditions of materials. Students learn to identify mechanical properties of all types of FRP (Fiber Reinforced Plastic) composite materials.
Facilities	Bottom pouring type stir casting machine, Universal testing machine, Digital ultrasonic flaw detector, Barcol hardness tester. Display boards provided for equipment list, experiments, timetables, and syllabus.
Used By	IV Year, I Semester

Computational Structural Analysis Lab

Purpose	Introduces overview of design and analysis of aircraft structural components. Students gain knowledge to design and analyse all different components of an aircraft, including model analysis on open and close sections.
Facilities	Relevant structural analysis software installed on all systems. White marker board and display boards provided.
Used By	IV Year, I Semester

9. Student Achievements

SAE Aero Design Challenge 2020

The Society of Automotive Engineers (SAE) conducted the 5th edition of the Aero Design Challenge 2020 from 29th February to 1st March 2020 at Bannari Amman Institute of Technology, Coimbatore. Total of 140 teams participated from across India. MLRIT teams bagged 3 prizes in various categories.

- Team VAYUPUTRA: 2nd Prize Overall (Cash Award: Rs. 50,000) and 1st Place for Best Design Report (Cash Award: Rs. 10,000) under Micro Class category.
- Team AIRAVAT: 3rd Place for Best Innovation under Regular Class category.
- Team Alpha: 3rd Best Overall Team award in Micro Class with a cheque of Rs. 25,000, presented by Shri K P Anbalagan, Honourable Minister of Higher Education, Tamilnadu.
- Team Sky Troopers: Best Design Report Award and cheque of Rs. 10,000 for the Regular Class, presented by Shri K P Anbalagan, Honourable Minister of Higher Education, Tamilnadu.
- Team Phoenix Flyers: 3rd Best Design Report award in Micro Class, presented by Shri K P Anbalagan, Honourable Minister of Higher Education, Tamilnadu.

Aircraft Build Competition (IAAA) 2019

MLRIT Aeronautical Engineering students participated in the Aircraft Build Competition (Design, Build and Fly) on 16 and 17 September 2019, organised by the Institute of Aeronautics Astronautics and Aviation (IAAA) at Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar. Five teams participated; four teams won prizes.

- Team Darksiders: Overall Championship.
- Team Black Bird: 1st Place for Best Design.
- Team Echo-19: 2nd Place for Technical Presentation.
- Team Thunderbird: 2nd Place in Build Team.

Shodhana-2017 National Level Competition

A. Harish (15R26D7601) and B. Prashanth (15R26D7608) of M.Tech Aerospace Engineering won the First Prize in the National Level Competition conducted by the National Institute of Rural Development and Panchayati Raj (NIRDPR) in association with ABVP, named Shodhana-

2017, at Rajendranagar, Hyderabad on 08/07/2017, for the project titled "Drone Pesticide Spraying Autonomous Agriculture Support". The first prize was presented by NITI Aayog member Dr. Saraswat, Central Minister Mr. Venkaiah Naidu, and Member of Parliament Mr. Konda Vishweshwara Reddy.

Boeing Externship Programme

- Mr. George Yuvraj and Ms. Shruthi (final year Aeronautical students) successfully completed their externship programme at Cyient-Boeing CEO, Hyderabad on 11 and 12 July 2016. They received certificates from Mr. Chakravarthy, Director, Boeing.
- CYIENT in collaboration with BOEING conducted an entry-level exam for III B.Tech Aeronautical Engineering students to select two students for a two-month summer externship. Ms. C. Sai Manish and Mr. K. Leela Krishna Sai Tarun were selected.

Dassault Systemes Recognition

- Nandini C., 4th Year Student, Department of Aeronautical Engineering, secured 6th place globally in "Best Project of the Year-2017" conducted by Dassault Systemes Academy.
- The department conducted a certification exam for present and alumni students on behalf of Dassault Systemes on 03/03/2017 by Mr. K. Shiva Shankar, Aero Department. 15 to 20 students attended; 90% qualified and received certification from Dassault Systemes.

Other Competitions and Events

- III B.Tech students M. Suman, Ch. Harish, G. Manasa, and C. Sai Manisha participated in the International Society for Scientific Research and Development (ISSRD) competition and presented a PPT on "Self Healing Materials in Aircraft Applications". They secured 1st position.
- IV year students Laharika and Sreedhar presented their EPIC project "TRICOPTER" for detection of crop diseases at the IEEE AIYEHUM CHALLENGE-2016 (IEEE Young Engineers Contest) held in July 2016 and secured 11th place among the world's top 20 projects.
- III B.Tech students Ch. Harish, G. Vamshi Krishna, and Aditya Tiwari participated in "Anveshana-2017" at Khairatabad, organised by Agastya Foundation on 21-23 January, presenting their project "BIO PAPER" made of vegetable wastes.
- Aeronautical Engineering students won the College Annual Sports Championship Award for 2016-2017.
- III year student Swarna Viswabrahmana (14R21A2154) was selected for the Volleyball Team at JNTUH.
- The department conducted Micro Projects for III and II B.Tech students. Management awarded cash prizes to the top three winners from each year. Prizes were distributed by Aeronautical Department Head Dr. Satyanarayana Gupta.

10. Student Internships

Academic Year 2022-23

S.No	Roll Number	Student Name	Organisation
1	19R21A2169	M Sushmitha	Tata Advanced Systems Limited
2	20R25A2103	G Shashank Netha	ASL DRDO
3	19R21A2182	U Lokesh	TIRVEN Industries Pvt. Ltd
4	19R21A2117	K Sindhuja	Keka Technologies Pvt Ltd

Academic Year 2023-24

S.No	Roll Number	Student Name	Organisation
1	20R21A2121	T Hari Hara Prasad	ARCI
2	20R21A2138	N Yashwanth	DRDO-DRDL
3	20R21A2143	P Vivek	BHEL
4	20R21A2108	B Sai Rohith	DYSL-AT-DRDO

Academic Year 2024-25

S.No	Roll Number	Student Name	Organisation
1	21R21A2116	Gone Shashank	Tata Lockheed Martin Aerostructures Ltd
2	21R21A2124	J Ajay Kumar Reddy	WINGS & PROPS PVT LTD
3	21R21A2125	Srenik Kumar	WINGS & PROPS PVT LTD
4	21R21A2152	U Abhinay Raj	WINGS & PROPS PVT LTD

Academic Year 2025-26

S.No	Roll Number	Student Name	Organisation
1	22R21A2101	A Vinay Teja	Tata Sikorsky Aerospace Limited
2	22R21A2155	J Shaik Sohail	Indian Institute of Technology, Hyderabad
3	22R21A2112	D Ramu	Tata Advanced Systems Limited

11. Special Programmes and Centres of Excellence

Centres of Excellence

- Product Lifecycle Management (PLM) — First ever in South India, established in collaboration with Tata Technologies.
- Flight Simulation — Established with assistance from AerX Labs India Pvt. Limited and CSIR-NAL.
- Digital Manufacturing
- Computational Engineering

Other Special Programmes

- CATIA Certification Centre by Dassault Systemes — the only such centre in Andhra Pradesh and Telangana.
- Boeing Externship Programme — two months paid externship (Rs. 10,000 per month) for two III B.Tech II Semester students.
- Special Diploma Course for employees of Tata Advanced Systems Limited (TASL).
- In-house Campus Recruitment Training (CRT) programme conducted throughout the year.

12. Professional Associations

- AeSI — Aeronautical Society of India
- SAE — Society of Automotive Engineers (Students participated in SAE International competition at Florida, USA)
- ISTE — Indian Society for Technical Education
- IAAA — Institute of Aeronautics, Astronautics and Aviation

13. Research and Publications

- Faculty and students have published approximately 160 papers in Scopus-indexed journals.
- Two books published in core subjects.
- 10 patents published in 2019-2020.

- DST-funded projects under FIST: Rs. 50 Lakhs worth of projects being handled.
- AICTE-funded project: Rs. 17 Lakhs.
- Received funding from ISRO and SERB to organise international conferences ICAAMM-2016 and ICAAMM-2018.

End of Document

Source: Aeronautical Engineering Department — MLR Institute of Technology